

SAFEAD AI (SAFE-VISION)

Multimodal Pre-Publication Advertisement Safety & Policy Moderation Framework

1. Executive Summary & Objective

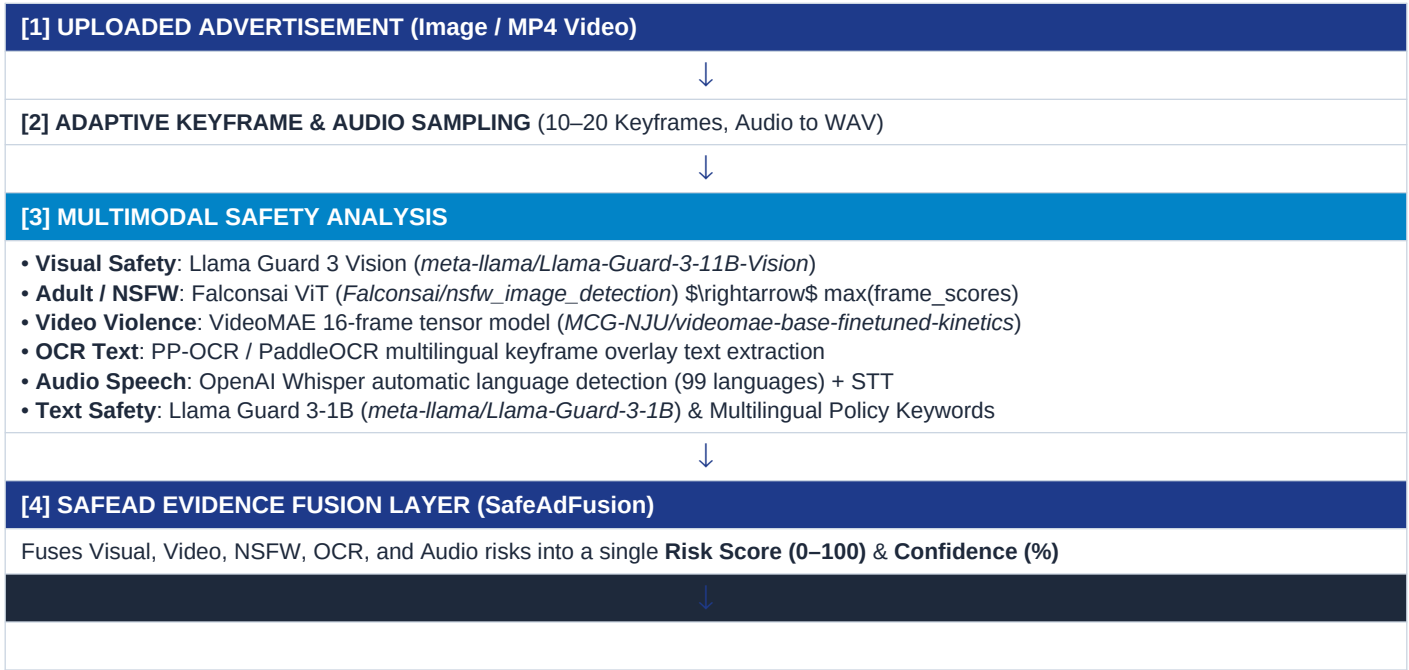
SafeAd AI is a pre-publication advertisement safety moderation framework. The system accepts uploaded image or video advertisements and determines whether they are permissible to publish, enforcing automated compliance across 5 safety modalities: **Visual Safety**, **Video Action Content**, **Embedded OCR Text**, **Audio Speech Transcription (Whisper)**, and **Contextual Retrieval**.

The final classification output maps deterministically into **exactly four policy classes**:

Target Class	Display Label	Policy Decision	Description
SAFE_FOR_ALL	Safe for All	APPROVED	Safe for general audience of all ages.
SAFE_14_PLUS	14+	AGE RESTRICTED 14+	Teen-restricted content (gambling/alcohol/tobacco).
SAFE_18_PLUS	18+	AGE RESTRICTED 18+	Adult-only non-prohibited content (mature themes).
UNSAFE_FOR_ALL	Unsafe for All	REJECTED	Severe violence, child risk, scams, or illegal acts.

2. System Architecture & Multimodal Pipeline Flow

The diagram below illustrates the end-to-end processing pipeline from advertisement ingestion to safety evidence fusion and final policy classification:



3. Pretrained Modality Detectors & Service Specifications

Modality	Pretrained Checkpoint	Key Responsibilities	Output Evidence Schema
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Visual Safety	Llama Guard 3 Vision (<i>meta-llama/Llama-Guard-3-11B-Vision</i>)	Evaluates keyframes for violence, sexual content, child safety, hate, and dangerous acts.	{category, unsafe: bool, confidence, source}
Adult / NSFW	Falconsai ViT (<i>Falconsai/nsfw_image_detection</i>)	Frame-level NSFW probabilities aggregated into video adult score via max(frame_scores).	{adult_score, adult_content_detected: bool, frame_scores: []}
Video Violence	VideoMAE Kinetics-400 (<i>MCG-NJU/vidomae-base-finetuned-kinetics</i>)	Direct (1, 16, 3, 224, 224) video action tensor classification for short and long ads.	{violence_score, violence_detected: bool, sampled_segments: []}
OCR Overlay	PP-OCR / PaddleOCR (Fallback: PyTesseract)	Multilingual keyframe text extraction, bounding boxes, confidence, and deduplication.	{ocr_text, ocr_confidence, frame_text: []}
Audio Speech	OpenAI Whisper (<i>openai/whisper-small / base</i>)	Auto-detects language across 99 languages, transcribes speech to text.	{audio_available: bool, language, transcript, audio_safety: {}}
Text Safety	Llama Guard 3-1B (<i>meta-llama/Llama-Guard-3-1B</i>)	Evaluates OCR text & speech transcripts against English, Hinglish, and Manglish keywords.	{violent, sexual, child_safety, gambling, overall_text_risk}

4. Project Directory & Package Structure

The project is structured into modular Python packages separation of concerns:

Package / Directory	Key Modules & Files	Functional Responsibility
processing/	- video_sampling.py - audio_extraction.py - preprocessing.py	Adaptive keyframe extraction (10-20 frames), PyAV audio stream detection, imageio_ffmpeg PATH binding.
models/	- model_manager.py - visual_safety.py - nsfw_detector.py - violence_detector.py - ocr_service.py - audio_service.py - text_safety.py	Pretrained safety detectors, sequential Google Colab VRAM memory flushing (torch.cuda.empty_cache), and inference services.
models/fusion/	safead_fusion.py	Deterministic SafeAd Fusion layer (SafeAdFusion) and policy decision engine.
backend/app/	- main.py - db/models.py, database.py - api/v1/moderation.py, advertisements.py	FastAPI REST backend endpoints (POST /api/moderate, GET /api/moderation/history) and MySQL database ORM.
frontend/	streamlit_app.py	Streamlit 8-step moderation workflow dashboard, action badges, risk scores, and User Feed.
tests/	test_moderation_pipeline.py	Automated integration test suite verifying 13 test scenarios.

5. Safety Priority Rules & Decision Hierarchy

The SafeAd Policy Engine enforces strict priority rules so that unsafe content is never downgraded to age-restricted or safe status:

- Child Exploitation / Severe Risk** $\rightarrow$ **UNSAFE_FOR_ALL (REJECT)**
- Severe Violent Action Content** $\rightarrow$ **UNSAFE_FOR_ALL (REJECT)**

- 3. **Misleading Ads / Scams / Illegal Acts** $\rightarrow$ **UNSAFE_FOR_ALL (REJECT)**
- 4. **Adult-Only Permissible Content** $\rightarrow$ **SAFE_18_PLUS (AGE RESTRICTED 18+)**
- 5. **Teen-Restricted Content (Gambling/Alcohol)** $\rightarrow$ **SAFE_14_PLUS (AGE RESTRICTED 14+)**
- 6. **General Audience Content** $\rightarrow$ **SAFE_FOR_ALL (APPROVED)**